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W8-03: ZooKeeper (and Chubby)

COMPSCI4064 (H) and COMPSCI5088 (M)

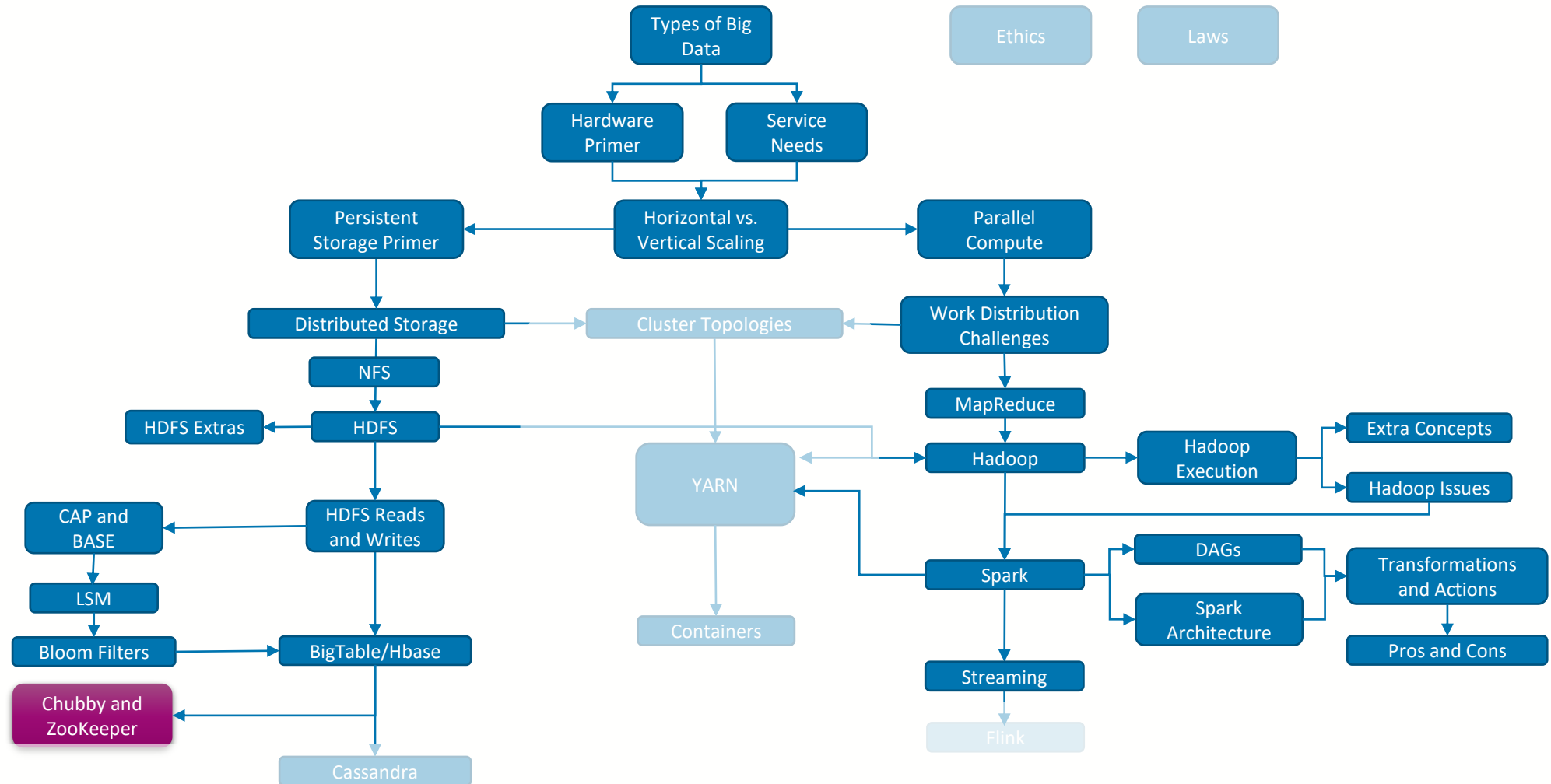
Dr. Richard McCreddie

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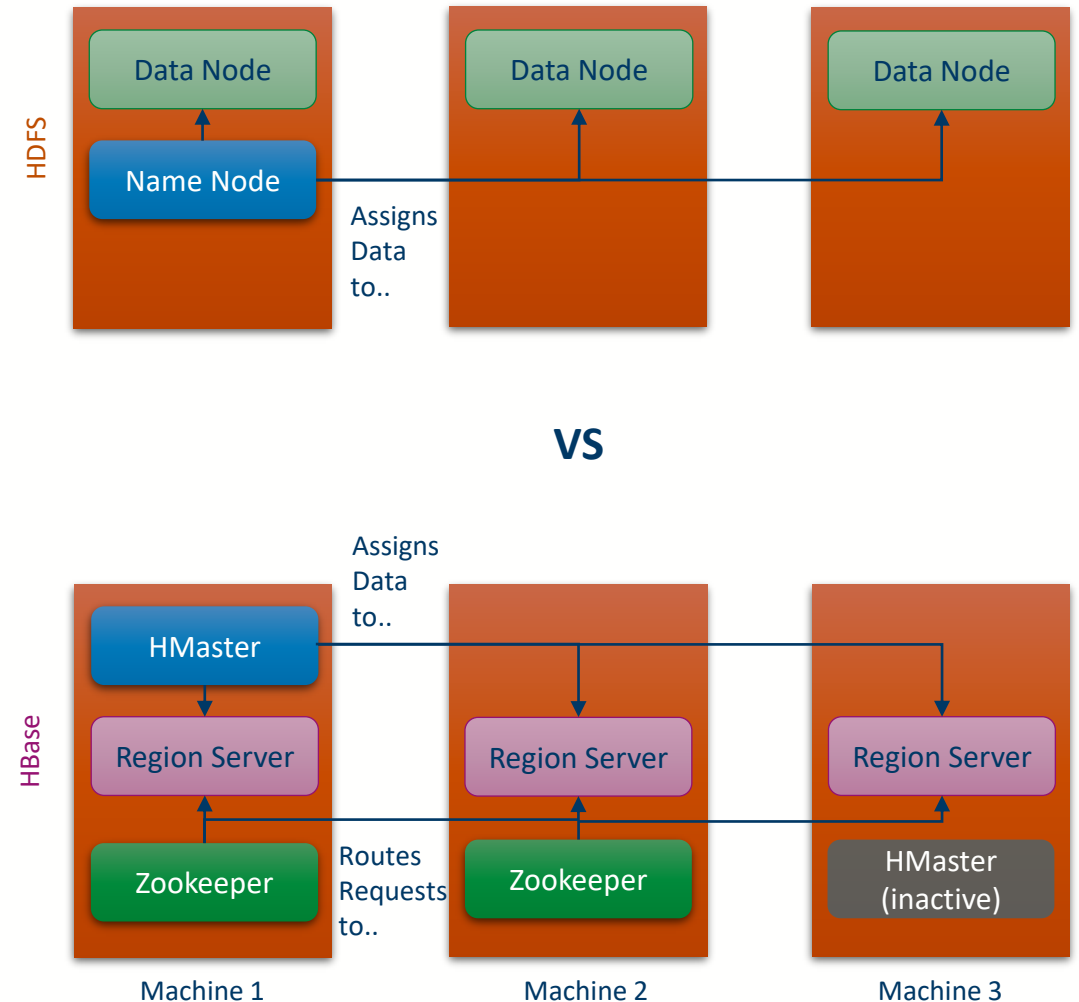
Where are we?





HBase Architecture

- HBase
 - **HMaster**: Assigns regions to Region Server. Creates/deletes tables. Routes data requests to region servers. Handle failures.
 - **Region Server**: Handles data requests from the user.
 - **Zookeeper**: Acts as a communication bridge. Tracks state of HMaster and Region Servers. Triggers recovery.





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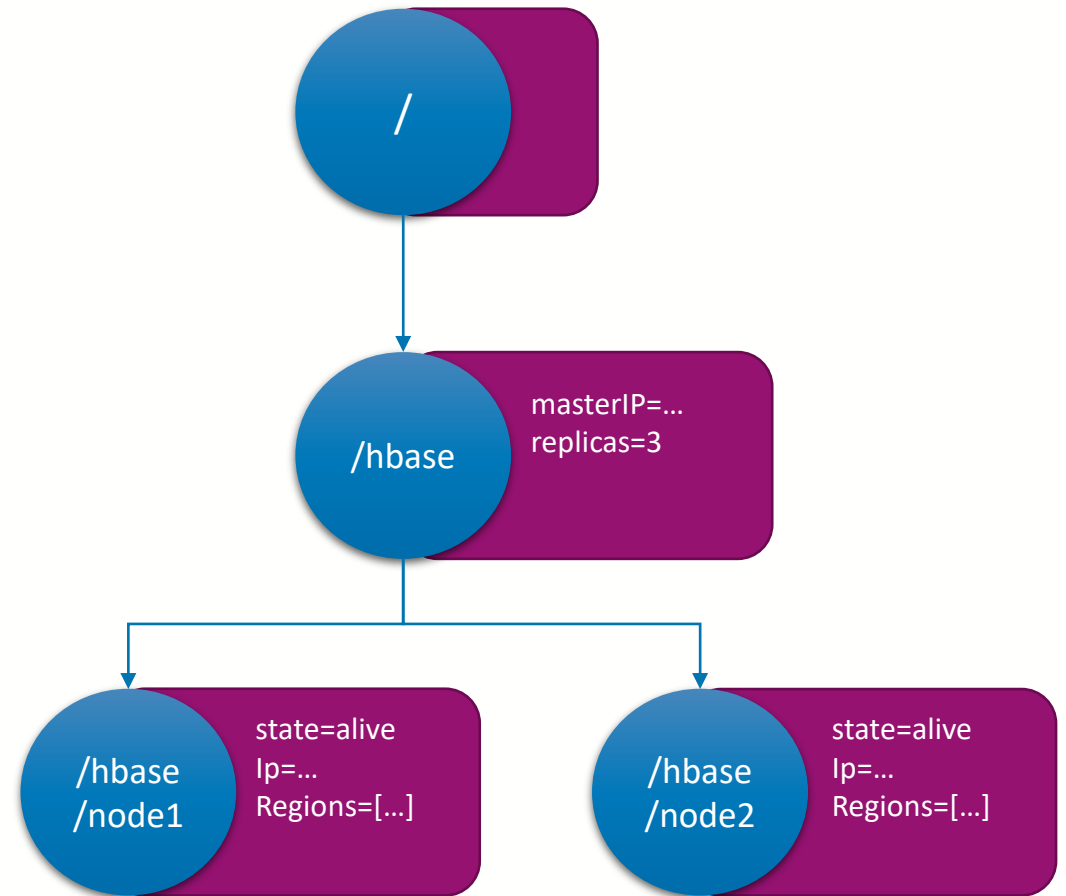
APACHE
ZooKeeper™

- ZooKeeper is a **distributed key-value data store** that provides the option for applications to be **notified if a value for a key changes**
 - Designed to be **lightweight and highly reliable**
- ZooKeeper is **everywhere**
 - Its not just an HBase service, also used by Spark, Flink
 - Used to power other things, e.g. Reddit, Twitter, Facebook/Meta, eBay...
- Based of the design of **Chubby** (another Google service)



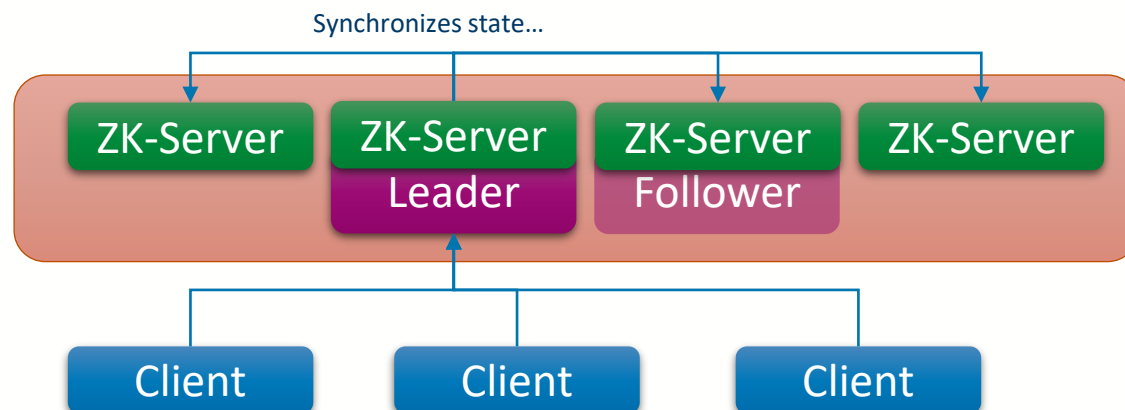
ZooKeeper Data Model

- ZooKeeper is a **key-value data store**, like a Map or Dictionary
 - Fundamentally simple
 - However, keys are structured into a hierarchal namespace
- Each node in the hierarchy is known as a **ZNode**
 - ZNodes hold a value which can be read or written to
 - Who can read/write is defined by an Access Control List (ACL)



Reliability through Replication

- ZooKeeper is comprised of **multiple** (usually at least 5) **servers**
 - One server is tagged as the **Leader**, with another the **Follower** (the next in line of the Leader fails to respond)
 - The Leader synchronizes its state with the other servers
- A distributed application runs a ZooKeeper Client within each of its services
 - Allows that application to connect to the leader to read and write data
 - Also regularly reports the application service's state to ZooKeeper





Why is ZooKeeper so Common?

- It is a solution to a common issue with distributed applications:
 - How do I track the state of the various replicas of those components?
- Applications that use ZooKeeper follow the school of thought that an application should not be responsible for tracking its own state, but should use an independent service for this
 - This avoids issues with 'master' services failing, bringing down the whole system
- All application components report their state to ZooKeeper
 - Management services within the application can ask ZooKeeper to notify them when data written by those components changes



Examples of using ZooKeeper

- **File Locking:** Hold the 'Lock' message for a file while it is being written
- **Store Global Configuration:** Store configuration variables or schemas of tables
- **Store File Access Controls:** For a file list hold the access permissions
- **Hold Bootstrap Configuration:** Store key start-up data, like where master services are located
- **Get the List of Service Replicas:** Have each service write down information about itself

HBase uses ZooKeeper for all of these!



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